

Entity-Relationship Diagram & Database Specification

Complete technical reference for the 10 core production entities and relational invariants

DATABASE ENGINE	CORE TABLES	NORMALIZATION	LEDGER CONSISTENCY
MySQL 8.0+ (InnoDB)	10 Active Entities	3NF / BCNF Complete	Audited Double-Entry Journal

1. Executive Architecture & Core Entities

MicroTeach implements an academic peer-tutoring marketplace and escrow settlement engine. Out of 16 database tables in the live schema, the platform runs on **10 active core tables**. The remaining 6 tables (e.g. legacy `Sessions`, standalone `Skills` / `User_Skills`, and abstract lookup helpers) are deprecated migration artifacts not queried by active runtime routes.

10 Production Core Entities

- Users:** Identity, roles, verification, and live wallet balances.
- Posts:** Student tutoring requests, deadlines, and bounties.
- Post_Applications:** Tutor applications, bids, and completion handshakes.
- Transactions:** Audited immutable financial journal and ledger.
- Teacher_Applications:** Academic credential validation pipeline.
- Reports:** Community content moderation queue.
- Transaction_Disputes:** Mediation and escrow refund arbitration.
- Reviews:** Bidirectional peer evaluation and feedback.
- Conversations & Messages:** Real-time sequential chat subsystem.

Filtered / Legacy Tables

- Sessions:** Deprecated legacy artifact; replaced by `Posts` + `Post_Applications`.
- Skills & User_Skills:** Unused standalone tables; expertise is handled in `Teacher_Applications` and course codes on `Posts`.
- Master_Transaction_Types:** Abstract enum cache; runtime types live directly in `Transactions.type`.
- Conversation_Members:** Auxiliary membership join table.
- Teacher_Application_Documents:** Sub-table for raw upload blobs.

2. Visual Entity-Relationship Diagram (10 Core Tables)

PK Primary Key **FK** Foreign Key **UK** Unique Key **||--o{** One-to-Many **||--o|** One-to-Zero-or-One

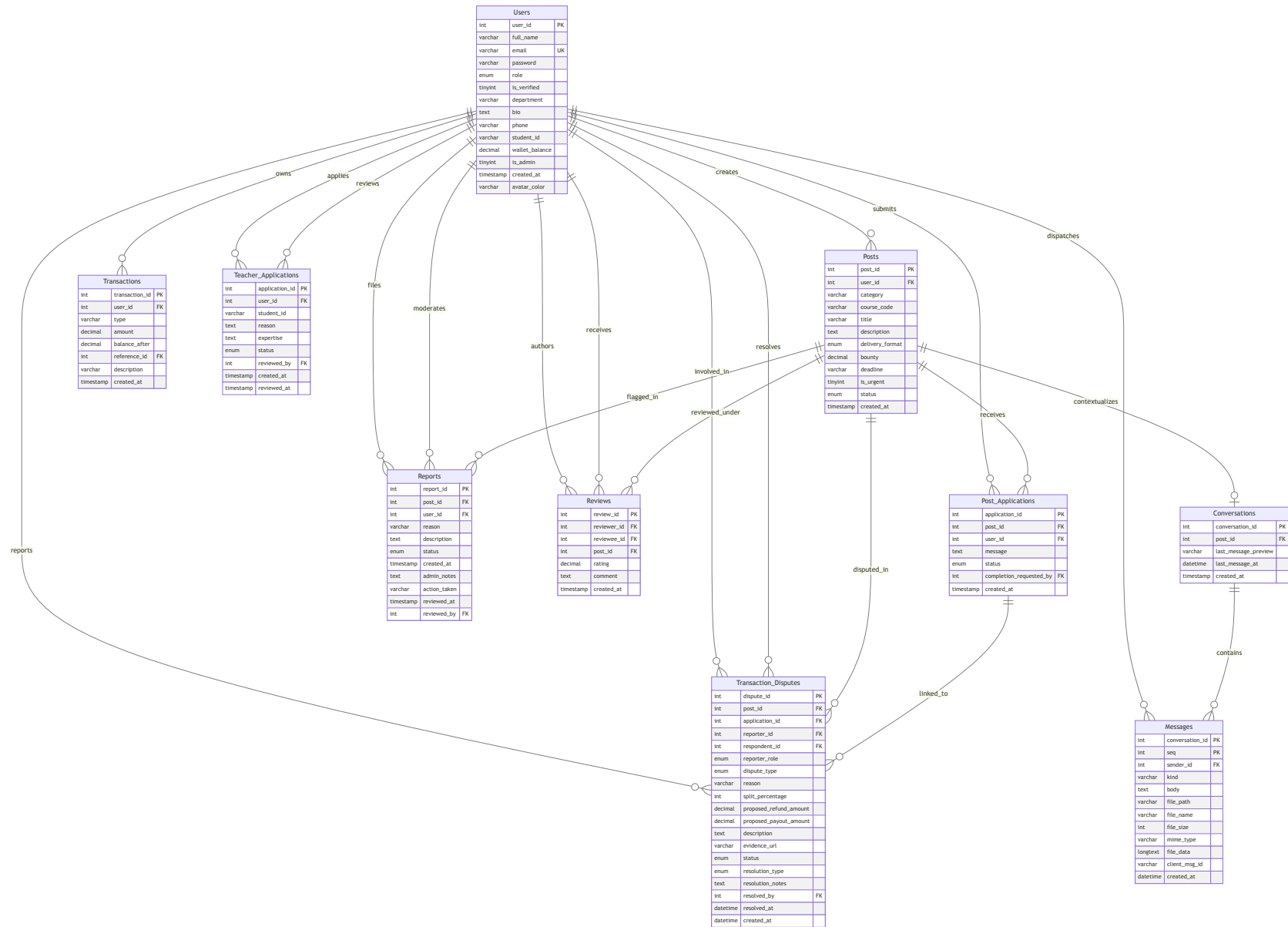


Figure 1.1: MicroTeach Full Relational Entity Architecture & Cardinalities

3. Entity DDL & Attribute Reference

1. Users Account identity, credentials, roles, and spendable balance					
COLUMN	TYPE	NULL	KEY	DEFAULT	DESCRIPTION & BUSINESS LOGIC
user_id	INT	NO	PK	AUTO_INC	Unique user identity identifier.
full_name	VARCHAR(100)	NO	—	—	User's legal / university display name.
email	VARCHAR(150)	NO	UK	—	Unique university email address (primary login credential).
password	VARCHAR(255)	NO	—	—	Bcrypt salted hash of user's authentication password.
role	ENUM('student','tutor','both')	YES	—	'student'	Role authorization level for marketplace participation.
is_verified	TINYINT(1)	YES	—	0	Verification badge indicating admin-approved credentials.
department	VARCHAR(100)	YES	—	NULL	Academic department (e.g., Computer Science, Engineering).
bio	TEXT	YES	—	NULL	User profile summary and academic interests.
phone	VARCHAR(20)	YES	—	NULL	Emergency contact telephone number.
student_id	VARCHAR(50)	YES	—	NULL	University student identification number.
wallet_balance	DECIMAL(10,2)	YES	—	0.00	Current spendable account balance (synchronized with Transactions).
is_admin	TINYINT(1)	YES	—	0	Flag granting administrative dashboard and moderation access.
created_at	TIMESTAMP	YES	—	CURRENT_TIMESTAMP	Account creation timestamp.
avatar_color	VARCHAR(7)	YES	—	NULL	Hex color code assigned for initials-based user avatar fallback.

2. Posts Tutoring requests, course tags, deadlines, and bounty amounts					
COLUMN	TYPE	NULL	KEY	DEFAULT	DESCRIPTION & BUSINESS LOGIC
post_id	INT	NO	PK	AUTO_INC	Unique tutoring post identifier.
user_id	INT	NO	FK	—	Authoring student (References Users.user_id).
category	VARCHAR(100)	NO	—	—	Subject category classification.
course_code	VARCHAR(50)	NO	—	—	Course identifier (e.g., CSE327, MAT120).
title	VARCHAR(120)	NO	—	—	Headline summary of the academic inquiry.
description	TEXT	YES	—	NULL	Detailed problem statement and expectations.
delivery_format	ENUM('live_call', 'annotated_pdf', 'video_walkthrough')	YES	—	'live_call'	Preferred delivery medium for the tutoring session.
bounty	DECIMAL(10,2)	YES	—	0.00	Monetary reward locked into platform escrow upon agreement.
deadline	VARCHAR(100)	YES	—	NULL	Target completion date/timeframe.
is_urgent	TINYINT(1)	YES	—	0	Visual flag for high-priority emergency tutoring tasks.
status	ENUM('active', 'pending', 'resolved', 'closed', 'completed')	YES	—	'active'	Lifecycle status of the marketplace posting.
created_at	TIMESTAMP	YES	—	CURRENT_TIMESTAMP	Publication timestamp.

3. Post_Applications Tutor proposals, acceptance, and dual-party completion handshakes					
COLUMN	TYPE	NULL	KEY	DEFAULT	DESCRIPTION & BUSINESS LOGIC
application_id	INT	NO	PK	AUTO_INC	Unique application identifier.
post_id	INT	NO	FK	—	Targeted tutoring request (References Posts.post_id).
user_id	INT	NO	FK	—	Applying tutor user (References Users.user_id).
message	TEXT	YES	—	NULL	Proposal pitch explaining qualifications and approach.
status	ENUM('pending', 'accepted', 'rejected', 'cancellation_requested', 'cancelled', 'completion_requested', 'completed')	YES	—	'pending'	Current progress and handshake state.
completion_requested_by	INT	YES	FK	NULL	User ID who initiated the completion handshake.
created_at	TIMESTAMP	YES	—	CURRENT_TIMESTAMP	Application submission timestamp.

4. Transactions			Audited double-entry financial ledger and balance snapshots		
COLUMN	TYPE	NULL	KEY	DEFAULT	DESCRIPTION & BUSINESS LOGIC
transaction_id	INT	NO	PK	AUTO_INC	Unique ledger entry identifier.
user_id	INT	NO	FK	—	Account holder affected (References Users.user_id).
type	VARCHAR(50)	NO	—	—	Transaction type (deposit, escrow_lock, escrow_release, refund, admin_adjustment).
amount	DECIMAL(10,2)	NO	—	0.00	Signed monetary sum transferred.
balance_after	DECIMAL(10,2)	NO	—	0.00	Immutable wallet balance recorded immediately post-transaction.
reference_id	INT	YES	FK	NULL	Associated entity ID (e.g. post_id, dispute_id).
description	VARCHAR(255)	YES	—	NULL	Human-readable ledger description.
created_at	TIMESTAMP	YES	—	CURRENT_TIMESTAMP	Immutable ledger timestamp.

5. Teacher_Applications			Tutor verification submissions and administrative approvals		
COLUMN	TYPE	NULL	KEY	DEFAULT	DESCRIPTION & BUSINESS LOGIC
application_id	INT	NO	PK	AUTO_INC	Unique teacher application identifier.
user_id	INT	NO	FK	—	Applying student (References Users.user_id).
student_id	VARCHAR(50)	YES	—	NULL	Institutional student ID for academic verification.
reason	TEXT	YES	—	NULL	Applicant statement and motivation.
expertise	TEXT	YES	—	NULL	Academic subjects, skills, and qualifications.
status	ENUM('pending','approved','rejected')	YES	—	'pending'	Current administrative review status.
reviewed_by	INT	YES	FK	NULL	Administrator who processed the application (References Users.user_id).
created_at	TIMESTAMP	YES	—	CURRENT_TIMESTAMP	Application submission timestamp.
reviewed_at	TIMESTAMP	YES	—	NULL	Adjudication timestamp.

6. ReportsCommunity content moderation tickets and triage decisions					
COLUMN	TYPE	NULL	KEY	DEFAULT	DESCRIPTION & BUSINESS LOGIC
report_id	INT	NO	PK	AUTO_INC	Unique moderation ticket identifier.
post_id	INT	NO	FK	—	Reported post (References Posts.post_id).
user_id	INT	NO	FK	—	Complainant user (References Users.user_id).
reason	VARCHAR(50)	NO	—	—	Categorical violation code (e.g. academic_dishonesty, spam, harassment).
description	TEXT	YES	—	NULL	Factual details submitted by the reporter.
status	ENUM('pending','reviewed','dismissed')	YES	—	'pending'	Moderation queue ticket status.
admin_notes	TEXT	YES	—	NULL	Internal moderator investigation notes.
action_taken	VARCHAR(50)	YES	—	NULL	Enforcement outcome (e.g. post_removed, warning_issued, none).
reviewed_by	INT	YES	FK	NULL	Reviewing administrator ID.
reviewed_at	TIMESTAMP	YES	—	NULL	Resolution timestamp.
created_at	TIMESTAMP	YES	—	CURRENT_TIMESTAMP	Filing timestamp.

7. Transaction_Disputes					
Escrow claim mediation, split payout percentages, and resolutions					
COLUMN	TYPE	NULL	KEY	DEFAULT	DESCRIPTION & BUSINESS LOGIC
dispute_id	INT	NO	PK	AUTO_INC	Unique dispute case identifier.
post_id	INT	NO	FK	—	Associated tutoring post (References Posts.post_id).
application_id	INT	YES	FK	NULL	Targeted engagement (References Post_Applications.application_id).
reporter_id	INT	NO	FK	—	Complainant (References Users.user_id).
respondent_id	INT	YES	FK	NULL	Opposing party (References Users.user_id).
reporter_role	ENUM('student','tutor')	NO	—	—	Role of party initiating the formal claim.
dispute_type	ENUM('full_refund','split','full_payment')	NO	—	—	Claimant's sought settlement structure.
reason	VARCHAR(150)	NO	—	—	Primary reason for claim.
split_percentage	INT	YES	—	50	Proposed refund share for student (0-100%).
proposed_refund_amount	DECIMAL(10,2)	YES	—	0.00	Calculated refund claim amount.
proposed_payout_amount	DECIMAL(10,2)	YES	—	0.00	Calculated tutor compensation amount.
description	TEXT	NO	—	—	Detailed factual argument and timeline.
evidence_url	VARCHAR(512)	YES	—	NULL	Uploaded screenshots, documents, or logs.
status	ENUM('pending','under_review','resolved','dismissed')	YES	—	'pending'	Adjudication pipeline status.
resolution_type	ENUM('full_refund','split','full_payment','dismissed')	YES	—	NULL	Binding administrative ruling.
resolution_notes	TEXT	YES	—	NULL	Official rationale for administrative resolution.
resolved_by	INT	YES	FK	NULL	Administrator who adjudicated the case.
resolved_at	DATETIME	YES	—	NULL	Case resolution timestamp.
created_at	DATETIME	YES	—	CURRENT_TIMESTAMP	Filing timestamp.

8. Reviews						Post-completion evaluations and numeric ratings
COLUMN	TYPE	NULL	KEY	DEFAULT	DESCRIPTION & BUSINESS LOGIC	
review_id	INT	NO	PK	AUTO_INC	Unique review identifier.	
reviewer_id	INT	NO	FK	—	User submitting rating (References <code>Users.user_id</code>).	
reviewee_id	INT	NO	FK	—	User receiving rating (References <code>Users.user_id</code>).	
post_id	INT	NO	FK	—	Completed tutoring post (References <code>Posts.post_id</code>).	
rating	DECIMAL(2,1)	NO	—	—	Numeric evaluation score (1.0 to 5.0).	
comment	TEXT	YES	—	NULL	Qualitative testimonial / feedback.	
created_at	TIMESTAMP	YES	—	CURRENT_TIMESTAMP	Submission timestamp.	

9. Conversations						Chat threads, contextual post linkages, and latest previews
COLUMN	TYPE	NULL	KEY	DEFAULT	DESCRIPTION & BUSINESS LOGIC	
conversation_id	INT	NO	PK	AUTO_INC	Unique conversation thread identifier.	
post_id	INT	YES	FK	NULL	Contextual post initiating the conversation.	
last_message_preview	VARCHAR(255)	YES	—	NULL	Denormalized preview snippet for quick sidebar rendering.	
last_message_at	DATETIME(3)	YES	—	NULL	High-precision timestamp of most recent communication.	
created_at	TIMESTAMP	YES	—	CURRENT_TIMESTAMP	Thread creation timestamp.	

10. Messages					
Ordered chat dispatches, attachment files, and client deduplication					
COLUMN	TYPE	NULL	KEY	DEFAULT	DESCRIPTION & BUSINESS LOGIC
conversation_id	INT	NO	PK FK	—	Owning conversation thread (Composite PK).
seq	INT	NO	PK	—	Monotonically increasing sequence number per conversation (Composite PK).
sender_id	INT	NO	FK	—	Authoring user (References Users.user_id).
kind	VARCHAR(30)	NO	—	'text'	Message classification (text, image, pdf, file).
body	TEXT	NO	—	—	Message textual content or media caption.
file_path	VARCHAR(512)	YES	—	NULL	Server disk storage path for uploaded file attachments.
file_name	VARCHAR(255)	YES	—	NULL	Original file name uploaded by user.
file_size	INT	YES	—	NULL	Attachment size in bytes.
mime_type	VARCHAR(100)	YES	—	NULL	Standard MIME type (e.g. image/jpeg, application/pdf).
file_data	LONGTEXT	YES	—	NULL	Inline base64 fallback data for optimistic previews.
client_msg_id	VARCHAR(64)	YES	—	NULL	Client UUID for idempotent delivery and optimistic UI updates.
created_at	DATETIME(3)	NO	—	CURRENT_TIMESTAMP(3)	Millisecond-precision dispatch timestamp.

4. Relational Workflows & System Invariants

- 4.1 Marketplace & Escrow Flow:** When a student publishes a tutoring task in `Posts`, the offered `bounty` is validated against `Users.wallet_balance`. When an application in `Post_Applications` is accepted, funds are escrowed via an audited negative balance entry in `Transactions`. Completion requires a mutual handshake: either party marks it complete (`completion_requested`), and upon counterparty confirmation, the post status becomes `completed` and escrow funds are disbursed to the tutor.
- 4.2 Ledger & Double-Entry Invariants:** No direct wallet mutations occur without a companion record in `Transactions`. The `Transactions.balance_after` field provides an immutable chronological audit trail that allows point-in-time reconstruction of any user's balance without replaying historical deltas.
- 4.3 Dispute Resolution & Split Payouts:** If tutoring quality or task delivery is contested, either party may file a `Transaction_Disputes` claim. The system allows proposal of full refunds, tutor payouts, or percentage splits (e.g. 50/50). Administrators arbitrate the dispute, and the resolution triggers automated compensatory ledger adjustments.
- 4.4 Real-Time Chat Sequencing:** The `Messages` table uses a composite primary key (`conversation_id`, `seq`). This ensures total ordering of messages without relying on distributed clock consistency across client devices. Denormalized fields on `Conversations` ensure high-performance $O(1)$ inbox queries.